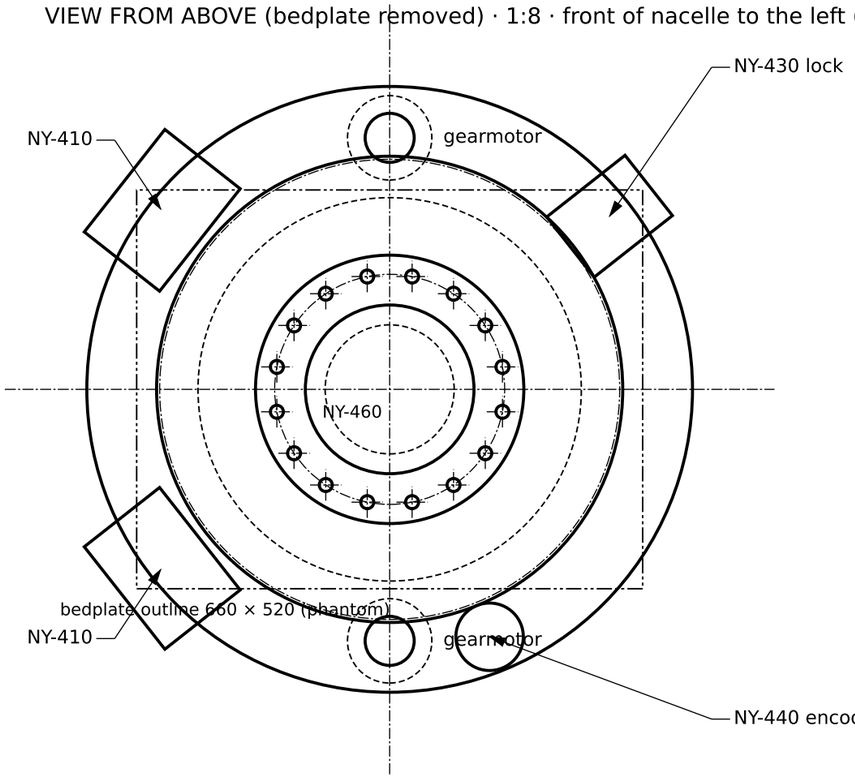
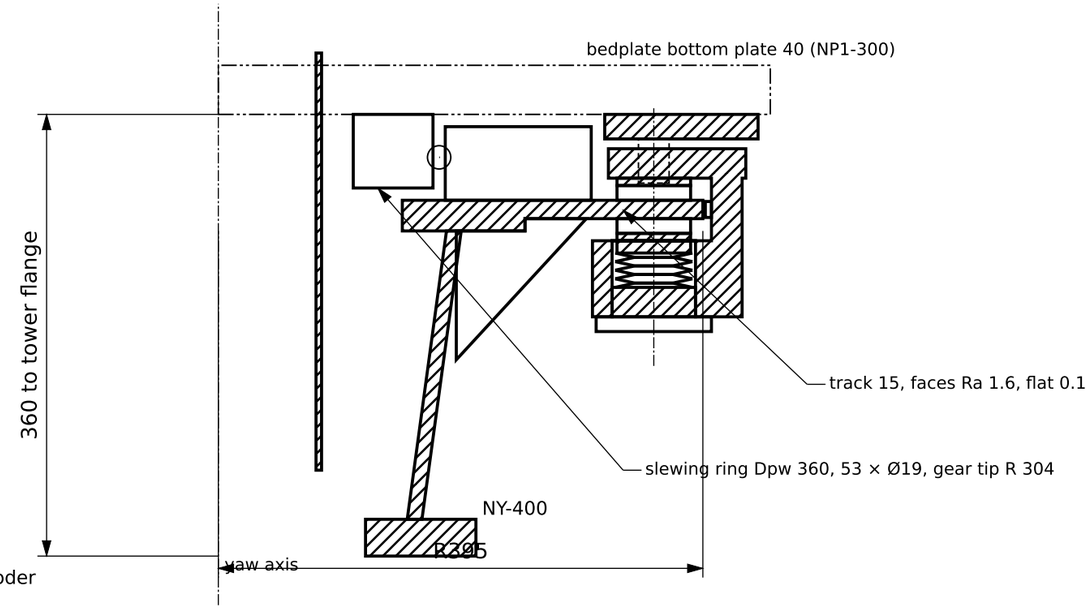
VIEW FROM ABOVE (bedplate removed) · 1:8 · front of nacelle to the left (−X)



SECTION A-A through the pad unit at 141.8° · 1:5



Item	Part	Qty	Make/buy	Mass
1	NY-400 tower-top adapter (S355J2+N weldment)	1	make	107 kg
2	Slewing ring Dpw 360, ext. gear z150 m4, ind. hardened teeth	1	buy	≈ 55 kg
3	Gearmotor 0.18 kW 4-pole, 1:1,680 planetary, 2 Nm motor brake	2	buy	≈ 22 kg
4	NP1-420 drive bracket (doc 16)	2	make	7.7 kg
5	NY-410 friction pad unit	2	make	13 kg
6	NY-430 yaw lock	1	make	6.5 kg
7	NY-440 encoder + cam limit switch unit	1	make/buy	6.9 kg
8	NY-460 cable hanger	1	make	10 kg
9	Bolts: 16 M16 inner, 16 M16 outer (from below), 16 M20 tower	48	buy	
10	Heated 2D ultrasonic + heated cup/vane on the nacelle mast	1+1	buy	

NOTES

1. No hydraulics on the yaw system. Holding: 2 spring-applied motor brakes (fail-safe) through the gear train; damping: 2 passive pad units, 2.4 to 3.6 kNm.
2. Design yaw moment held 8 kNm by ONE pinion (backlash) without friction credit; gearbox static rating ≥ 1,500 Nm output.
3. All bolts 10.9, preload with calibrated torque, M16 90 kN, M20 110 kN. Grease the ring per supplier (−40 °C synthetic).
4. Yaw rate 0.5 °/s at 1,500 rpm. One VFD drives both motors; the safety chain opens the brake supply directly.

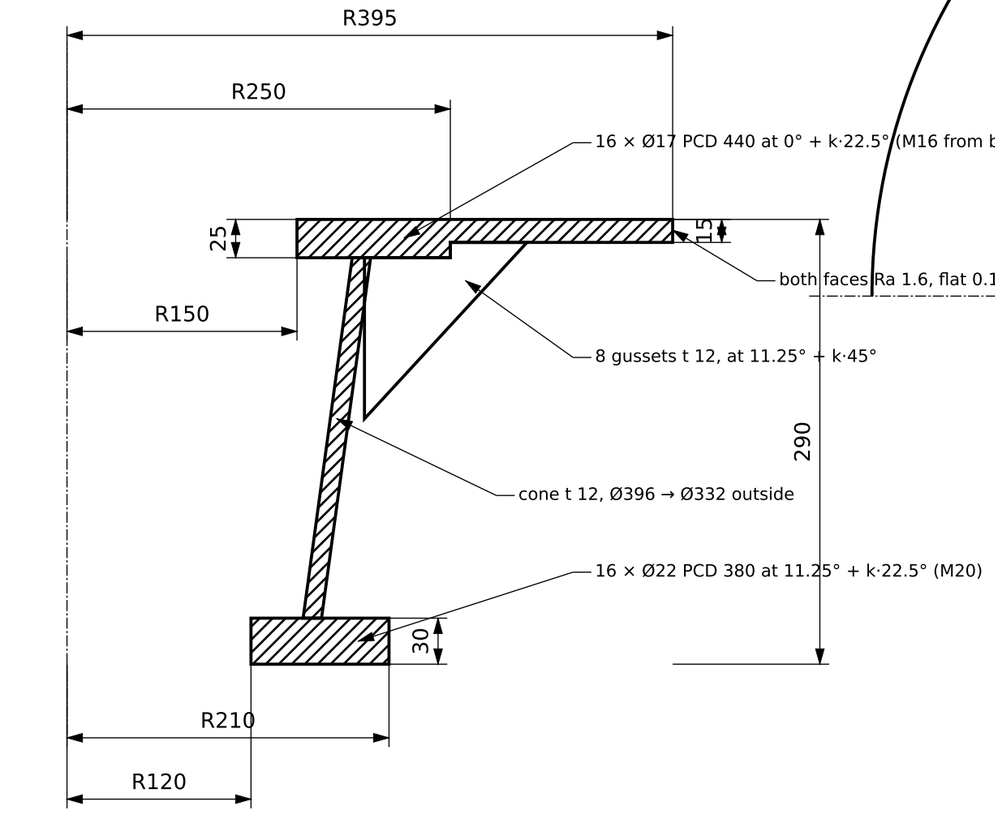
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

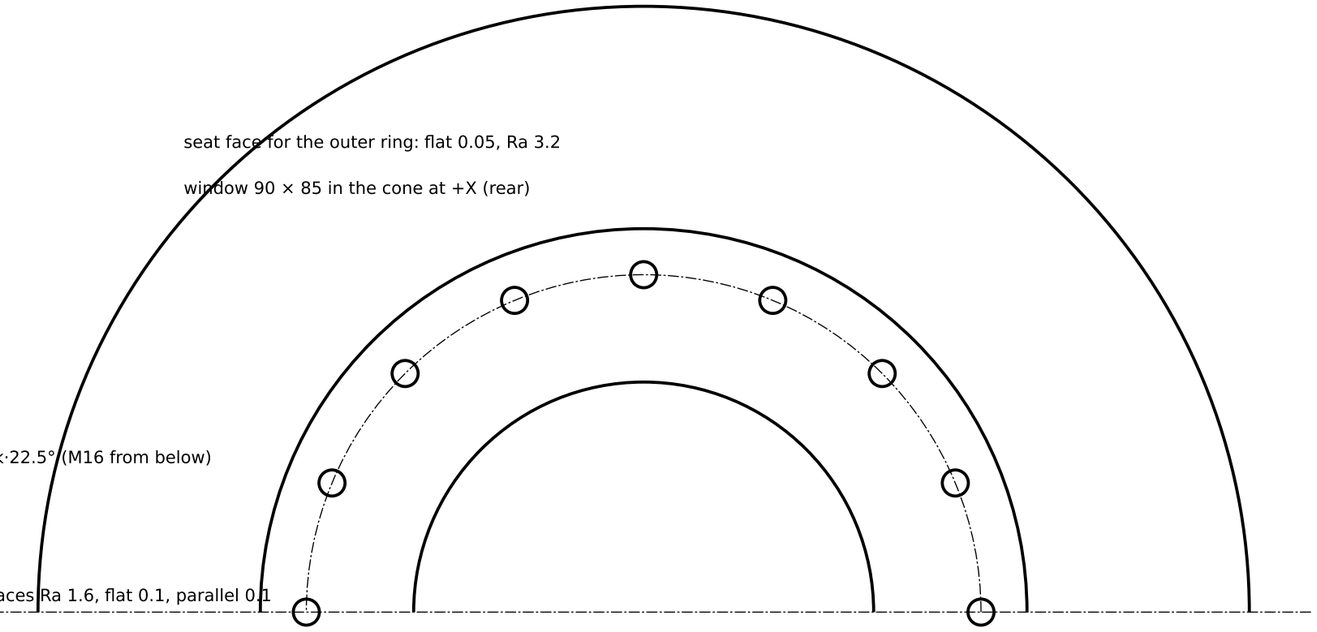
YAW SYSTEM · GENERAL ARRANGEMENT

DRAWING NO.	REV / DATE	SCALE (A3)
NY-000	A · 2026-11-05	1:5 / 1:8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see parts list	≈ 275 kg (yaw incl. adapter)	Claude / not checked

HALF SECTION (between gussets) · 1:4



PLAN FROM ABOVE · 1:4



NOTES

1. Weld cone to seat plate and to flange with full-penetration welds (ISO 5817 B), gussets with 5 mm double fillets; stress relieve, then machine seat, track faces and flange in one set-up.
2. Seat and track are one machined plate: 25 mm blank, track turned to 15 mm (r 250 to 395). Track faces left bare under the pads; all other surfaces C5-VH coating (ISO 12944, zinc-rich primer).
3. Bolt loads at the SLM tilt moment 34.6 kNm: outer ring 19.1 kN per M16 (preload 90 kN), tower flange 32.0 kN per M20 (preload 110 kN).
4. Replaces the NP1-400 proposal of doc 16 (25 mm plate to Ø840, 143 kg): track moved in to r 395 and thinned to 15 mm because the pads only clamp 5.3 kN.
5. Gussets stay inside r 300 and below the track so the lower pad jaws (r ≥ 318) pass all the way round.

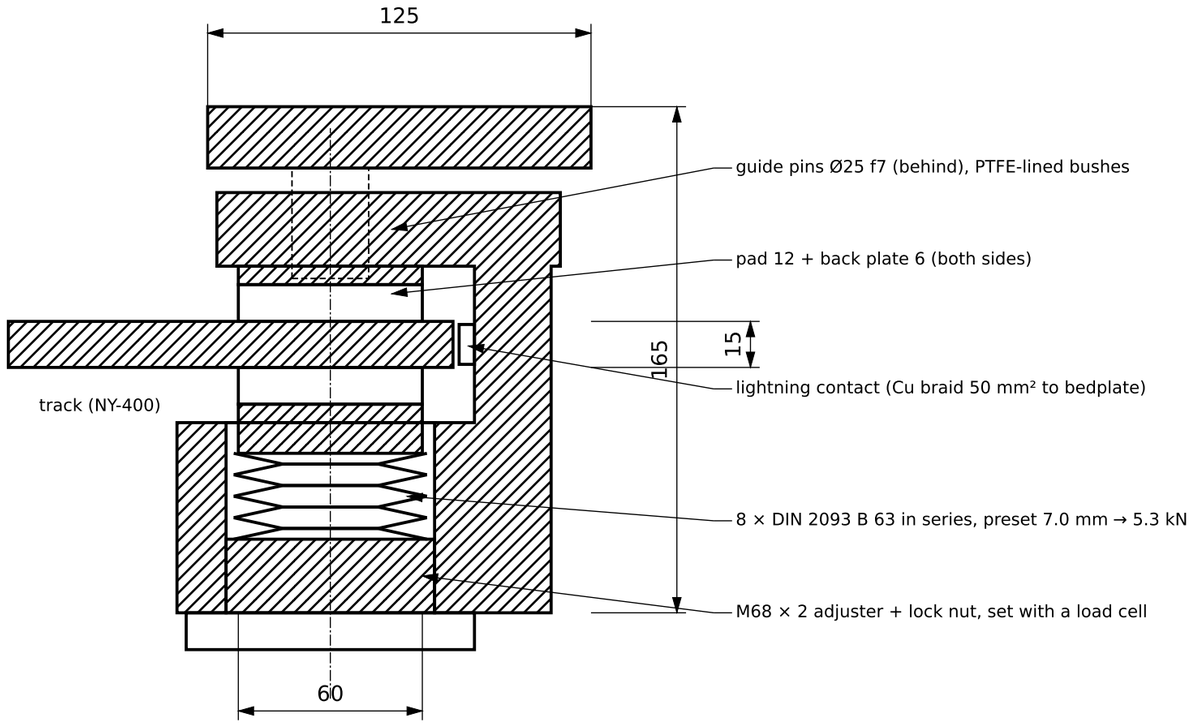
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

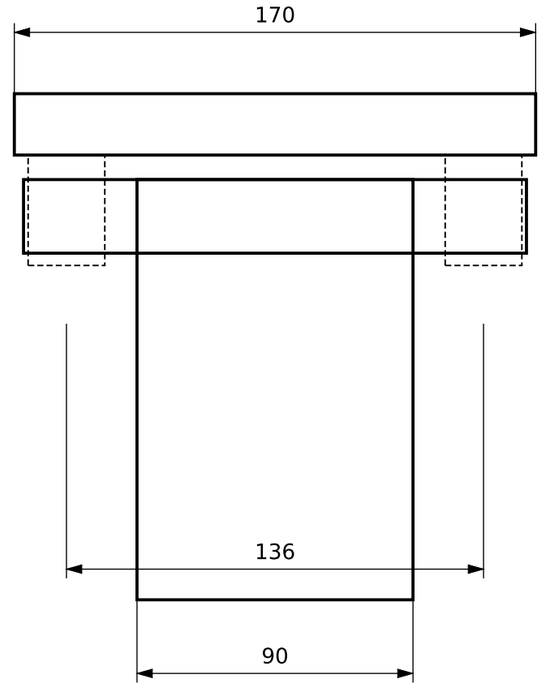
TOWER-TOP ADAPTER WITH FRICTION TRACK

DRAWING NO.	REV / DATE	SCALE (A3)
NY-400	A · 2026-11-05	1:4
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N plate and tube, weldment	107 kg (model)	Claude / not checked

SECTION IN THE RADIAL PLANE · 1:2



VIEW FROM OUTSIDE (radial) · 1:2



Item	Part	Qty	Material / spec
1	Bracket 170 × 125 × 20, 4 × Ø17 to the bedplate	1	S355J2+N
2	C-frame (upper jaw, web, lower jaw), machined	1	S355J2+N
3	Guide pin Ø25 f7 × 55, pressed into item 1	2	1.4462 duplex
4	Pad 90 × 60 × 12 on 6 mm back plate	2	organic industrial pad, μ 0.40 ± 20 %
5	Piston Ø60 × 10	1	1.4462
6	Disc spring DIN 2093 B 63 (63 × 31 × 2.5)	8	51CrV4, zinc-flake
7	Adjuster M68 × 2 + lock nut	1+1	1.4462
8	Lightning contact block + Cu braid 50 mm²	1	CuSn / Cu

NOTES

1. Floating C-frame: the springs push the lower pad up; the reaction pulls the frame down onto the upper pad, so both faces clamp 5.3 kN at any pad wear.
2. Two units give 2.4 to 3.6 kNm (μ 0.32 to 0.48), 3.0 kNm nominal, 1.51 × the 95th percentile operating yaw moment.
3. Pad pressure 0.98 MPa, sliding speed 3.1 mm/s, heat 26 W. Estimated wear 6.6 mm in 20 years of 9 mm usable (to confirm by test).
4. Spring stack at 0.50 h0 per disc: one year of wear lowers the clamp by 8 %. Re-set at the yearly service (load cell between the pads).
5. Pad change without lifting the nacelle: back off the adjuster, swing the frame down on the pins, slide the pads out sideways.

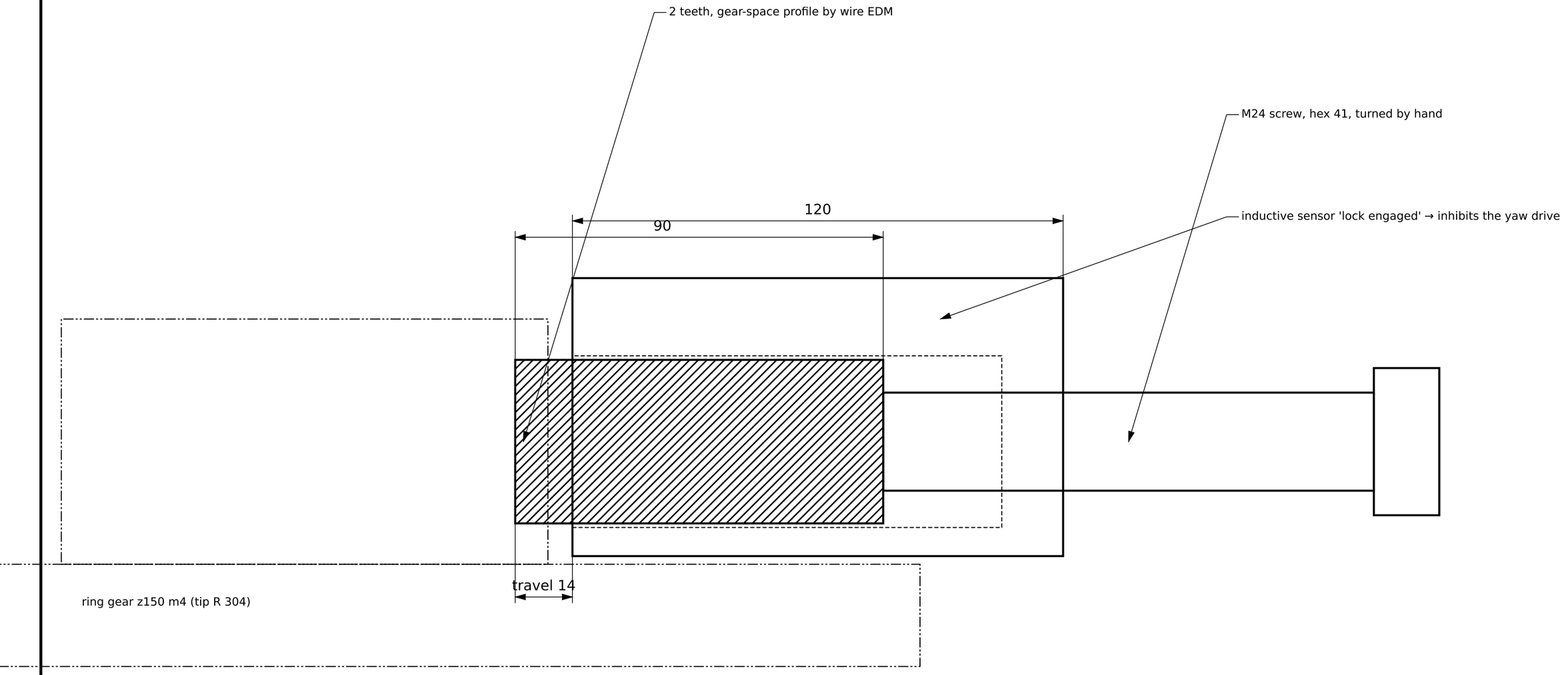
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

FRICTION PAD UNIT (PASSIVE, SPRING-LOADED)

DRAWING NO.	REV / DATE	SCALE (A3)
NY-410	A · 2026-11-05	1:2
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
frame S355J2+N, pins 1.4462	13 kg (model)	Claude / not checked

SECTION IN THE RADIAL PLANE (engaged) · 1:1



NOTES

1. Use: service only (yaw drives locked out). Turn the nacelle until a tooth space is opposite the block (encoder target), then screw the block in.
2. Strength: design yaw moment 8 kNm → 26.7 kN at the gear pitch radius on two teeth; tooth root shear and bending of the block S ≥ 3 (42CrMo4+QT, 900 MPa).
3. The housing bolts to the bedplate underside with 4 × M16 at the rear left corner (38.2°). Grease the slide; boot over the screw.

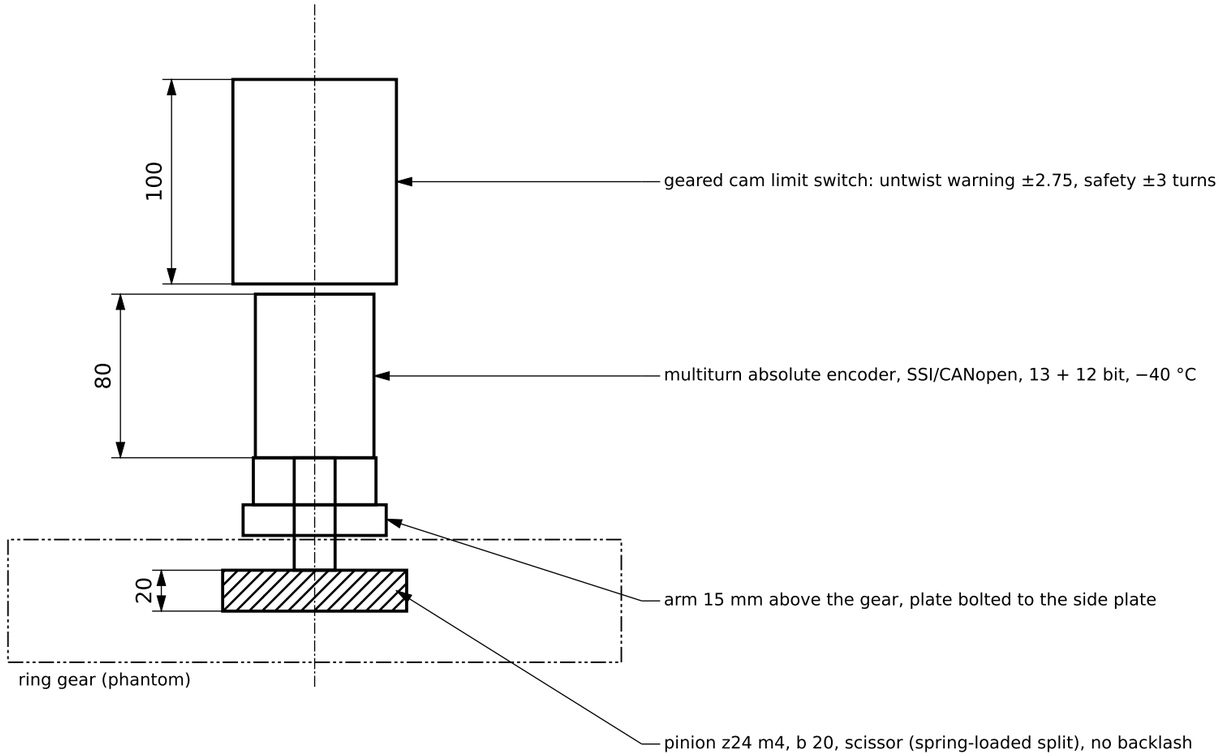
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

YAW LOCK (SERVICE)

DRAWING NO.	REV / DATE	SCALE (A3)
NY-430	A · 2026-11-05	1:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
housing S355J2+N, block 42CrMo4+QT	6.5 kg (model)	Claude / not checked

ELEVATION, VIEW TOWARDS THE SIDE PLATE · 1:3



NOTES

1. Ratio ring/pinion 150/24 = 6.25: ±3 nacelle turns = ±18.75 turns at the switch.
2. Centre distance 348 mm at -68° (right side, behind the drive), outside the side plate so the instruments stand beside it.
3. Cam contacts are positively opened, wired into the hardwired safety chain (PL d). The encoder is only for control (yaw angle, twist count, untwist).
4. Plausibility: proximity switch at 0° on a ring tooth marker checks the encoder every revolution.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

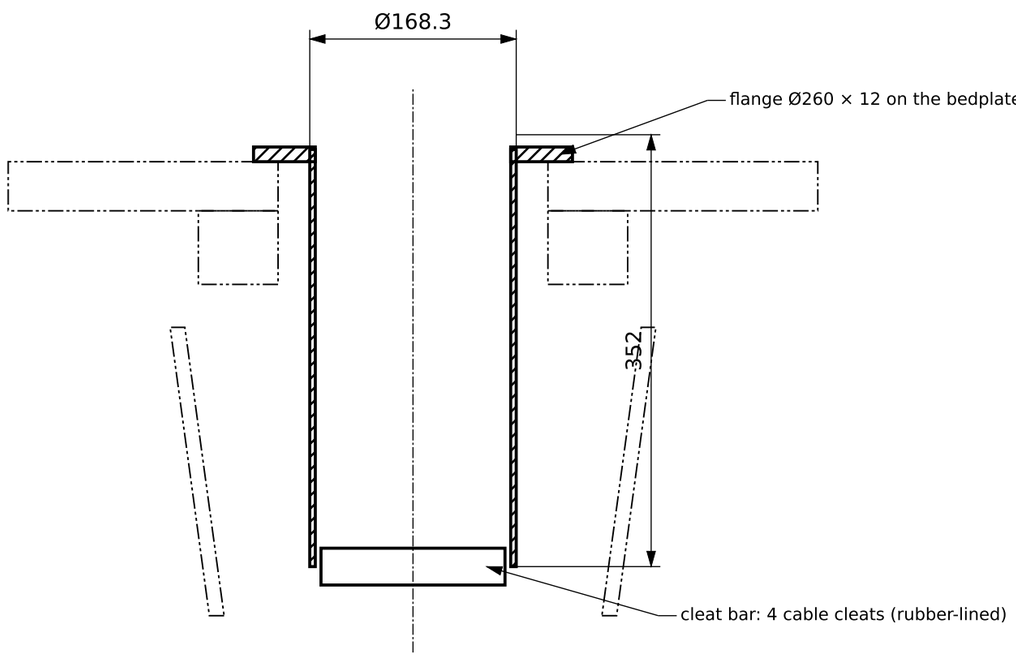
12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

YAW ENCODER AND CAM LIMIT SWITCH UNIT

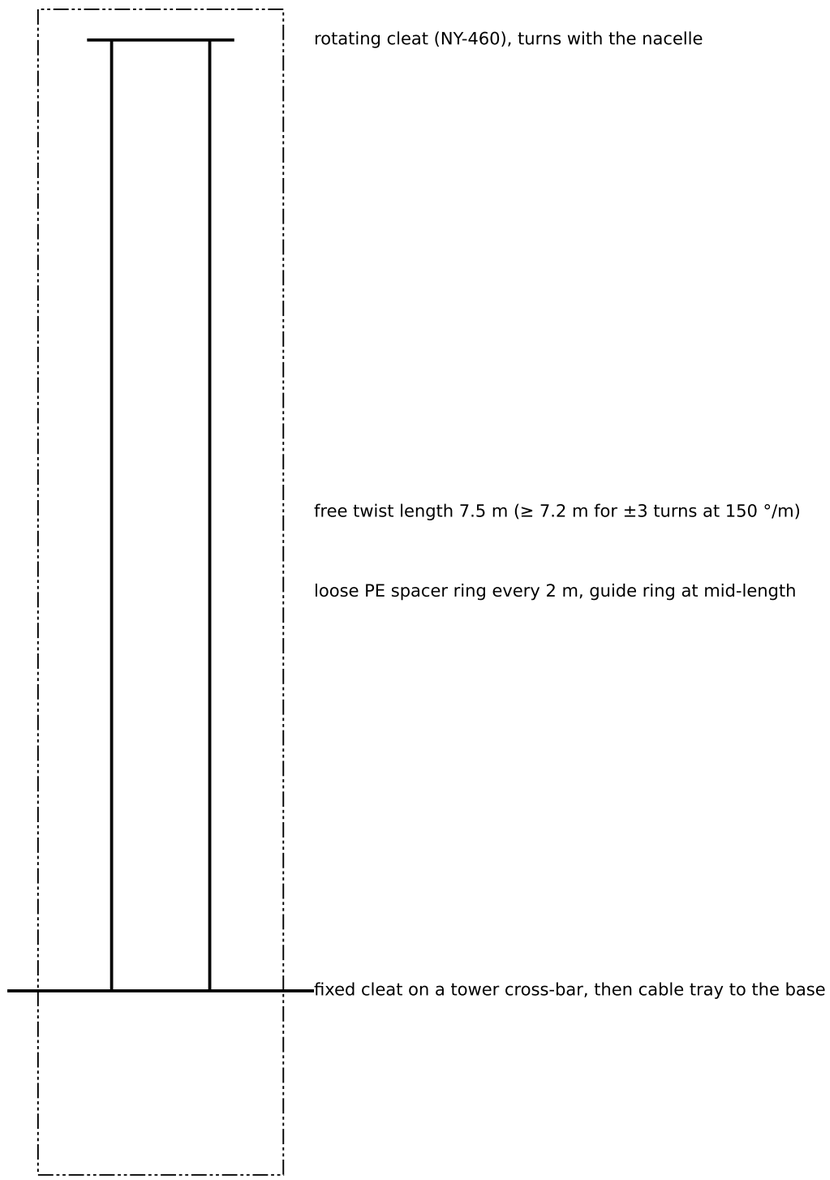
DRAWING NO.	REV / DATE	SCALE (A3)
NY-440	A · 2026-11-05	1:3
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
bracket S355J2+N, pinion 42CrMo4+QT	6.9 kg incl. instruments	Claude / not checked

Cable (all torsion-rated for wind turbine loops, −40 °C)	Ø approx.
Generator 4G6 + screen, 0.6/1 kV (doc 12)	20 mm
48 V auxiliary / backup supply 2 × 10 mm²	14 mm
Safety chain + control signals 12 × 1.0 mm², screened	12 mm
Industrial Ethernet (Cat 6A, torsion) + fibre option	12 mm
Earthing: separate Cu braid via the yaw lightning contact	–

HANGER IN THE YAW BORE · 1:5



TWIST LOOP IN THE TOWER (not to scale)



NOTES

1. Untwist by the controller at ±2.5 turns when the 10 min mean wind is below 5 m/s; safety chain opens at ±3 turns (NY-440 cam switch).
2. No slip ring for power (doc 07). Cables leave the bedplate inside the hanger tube, so they never touch the rotating ring or the adapter.
3. Inspection through the adapter window (NY-400) at every service: sheath wear, cleats, twist count against the encoder.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

CABLE HANGER AND TWIST LOOP

DRAWING NO.	REV / DATE	SCALE (A3)
NY-460	A · 2026-11-05	1:5 / NTS
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
tube S355 Ø168.3 × 4.5, flange S355J2+N	10 kg (model)	Claude / not checked